0.1 Solution

We place triangle ABC in the Cartesian plane. Let $B = (0, 0)$, $C = (17, 0)$. From $AB = 26$ and $AC = 25$ we obtain $A = (10, 24)$ (solving $x^2 + y^2 = 676$ and $(x - 17)^2 + y^2 = 625$ gives $x = 10,\ y = 24$).

0.1.1 1. Orthocenter H

The altitude from A is the vertical line $x = 10$. The altitude from B is perpendicular to AC. Since $\overrightarrow{AC} = (7, -24)$, a direction vector for this altitude is $(24, 7)$. Hence the line through B is $\{(24t, 7t)\}$. Intersecting with $x = 10$ gives $24t = 10 \Rightarrow t = 5/12$, so

$$H = \left(10,\ \frac{35}{12}\right).$$

0.1.2 2. Circumcircle Ω

The circumcenter O is the intersection of the perpendicular bisectors. The midpoint of BC is $(8.5, 0)$; the perpendicular bisector of BC is $x = 8.5$. The midpoint of AB is $(5, 12)$; the perpendicular bisector of AB has equation $2(x - 5) + 3(y - 12) = 0$ (since slope of AB is $24/10 = 12/5$, so perpendicular slope is $-5/12$). Solving with $x = 8.5$ yields $O = \left(\frac{17}{2},\ \frac{253}{24}\right)$. The radius is

$$R = |OA| = \sqrt{\left(10 - \frac{17}{2}\right)^2 + \left(24 - \frac{253}{24}\right)^2} = \frac{325}{24}.$$

0.1.3 3. Feet D, E, F and the nine-point circle

- D is the foot from A onto BC: $D = (10, 0)$.
- For E (foot from B onto AC): Project B onto line AC. With $A = (10, 24)$, $\overrightarrow{AC} = (7, -24)$, $|AC|^2 = 625$. The parameter t such that $A + t\overrightarrow{AC}$ is the foot is

  $$t = \frac{(B - A) \cdot \overrightarrow{AC}}{|AC|^2} = \frac{(-10, -24) \cdot (7, -24)}{625} = \frac{506}{625}.$$

  Hence

  $$E = A + t\overrightarrow{AC} = \left(10 + \frac{7 \cdot 506}{625},\ 24 - \frac{24 \cdot 506}{625}\right) = \left(\frac{9792}{625},\ \frac{2856}{625}\right).$$

- For F (foot from C onto AB): $\overrightarrow{AB} = (-10, -24)$, $|AB|^2 = 676$. Parameter s for the foot from C is

  $$s = \frac{(C - A) \cdot \overrightarrow{AB}}{|AB|^2} = \frac{(7, -24) \cdot (-10, -24)}{676} = \frac{506}{676} = \frac{253}{338}.$$

  Then

  $$F = A + s\overrightarrow{AB} = \left(10 - \frac{10 \cdot 253}{338},\ 24 - \frac{24 \cdot 253}{338}\right) = \left(\frac{425}{169},\ \frac{1020}{169}\right).$$

The circle through D, E, F is the **nine-point circle** of $\triangle ABC$. Its centre is the midpoint of O and H:

$$N = \left(\frac{\frac{17}{2} + 10}{2},\ \frac{\frac{253}{24} + \frac{35}{12}}{2}\right) = \left(\frac{37}{4},\ \frac{323}{48}\right),$$

and its radius is $R/2 = 325/48$.

0.1.4 4. Point X and the second intersection Q of AX with Ω

- Line EF: through E and F. Its intersection with BC (the x-axis) is found by setting $y = 0$ in the equation of EF. After straightforward algebra we obtain

$$X = \left(\frac{170}{3},\ 0\right).$$

- The line AX is parametrised as $A + \lambda v$ with $v = X - A = \left(\frac{140}{3},\ -24\right)$. The circumcircle Ω has equation $|P - O|^2 = R^2$. Substituting $P = A + \lambda v$ and using $|A - O|^2 = R^2$ gives a quadratic in λ:

$$|v|^2\lambda^2 + 2(A - O) \cdot v\,\lambda = 0,$$

  so the non-zero root is

$$\lambda_Q = -\frac{2(A - O) \cdot v}{|v|^2}.$$

  Compute

$$A-O = \left(10 - \frac{17}{2},\ 24 - \frac{253}{24}\right) = \left(\frac{3}{2},\ \frac{323}{24}\right), \qquad (A-O)\cdot v = \frac{3}{2}\cdot\frac{140}{3} + \frac{323}{24}\cdot(-24) = 70-323 = -253.$$

  Also

$$|v|^2 = \left(\frac{140}{3}\right)^2 + (-24)^2 = \frac{19600}{9} + 576 = \frac{24784}{9}.$$

  Hence

$$\lambda_Q = -\frac{2(-253)}{24784/9} = \frac{506 \cdot 9}{24784} = \frac{2277}{12392}.$$

  Therefore

$$Q = A + \lambda_Q v = \left(10 + \frac{2277}{12392} \cdot \frac{140}{3},\ 24 + \frac{2277}{12392} \cdot (-24)\right) = \left(\frac{57545}{3098},\ \frac{30345}{1549}\right).$$

0.1.5 5. Point P on the nine-point circle

A classical fact: the nine-point circle is the image of the circumcircle under the homothety with centre H and factor $\frac{1}{2}$. Consequently, for any line through H, the two intersection points of that line with the nine-point circle are exactly the midpoints of the segments from H to the two intersection points of the line with the circumcircle.

The line HQ meets Ω at Q and at another point (which we do not need explicitly). Hence the two intersections of HQ with the nine-point circle are the midpoints of H and those two circumcircle points. In particular, the midpoint of HQ lies on the nine-point circle and, because it lies between H and Q, it must be the point P described in the problem. Thus

$$P = \frac{H + Q}{2}.$$

0.1.6 6. Computation of the distances PQ and PE

We already have

$$H = \left(10,\ \frac{35}{12}\right), \qquad Q = \left(\frac{57545}{3098},\ \frac{30345}{1549}\right), \qquad E = \left(\frac{9792}{625},\ \frac{2856}{625}\right).$$

First compute HQ. Let $w = A - H = \left(0,\ \frac{253}{12}\right)$. From the expression for Q we have $Q - H = w + \lambda_Q v$. Using $\lambda_Q = \dfrac{506}{|v|^2}$ and the previously found $w \cdot v = -506$, we obtain

$$|Q - H|^2 = |w|^2 + 2\frac{506}{|v|^2}(w \cdot v) + \frac{506^2}{|v|^2} = |w|^2 - \frac{506^2}{|v|^2}.$$

Now

$$|w|^2 = \left(\frac{253}{12}\right)^2 = \frac{64009}{144}, \qquad |v|^2 = \frac{24784}{9}.$$

Thus

$$\frac{506^2}{|v|^2} = \frac{256036 \cdot 9}{24784} = \frac{2304324}{24784} = \frac{576081}{6196}.$$

Hence

$$|Q - H|^2 = \frac{64009}{144} - \frac{576081}{6196} = \frac{78411025}{223056}.$$

Therefore

$$HQ = \sqrt{\frac{78411025}{223056}} = \frac{8855\sqrt{1549}}{18588}.$$

Since P is the midpoint of HQ,

$$PQ = \frac{1}{2}HQ = \frac{8855\sqrt{1549}}{37176}.$$

Next compute PE. Because $P = \frac{H+Q}{2}$,

$$PE^2 = \left|\frac{H+Q}{2} - E\right|^2 = \frac{1}{4}\,|H + Q - 2E|^2.$$

After substituting the coordinates and simplifying (a lengthy but straightforward calculation) we obtain

$$PE = \frac{161161\sqrt{1549}}{929400}.$$

0.1.7 7. The ratio PQ/PE

$$\frac{PQ}{PE} = \frac{\dfrac{8855\sqrt{1549}}{37176}}{\dfrac{161161\sqrt{1549}}{929400}} = \frac{8855 \cdot 929400}{37176 \cdot 161161}.$$

Factor each number:

- $8855 = 5 \cdot 7 \cdot 253 = 5 \cdot 7 \cdot 11 \cdot 23$.
- $929400 = 600 \cdot 1549 = 2^3 \cdot 3 \cdot 5^2 \cdot 1549$.
- $37176 = 24 \cdot 1549 = 2^3 \cdot 3 \cdot 1549$.
- $161161 = 7^2 \cdot 11 \cdot 13 \cdot 23$.

Cancelling the common factors 1549, 11, 23, and the factor 24 from the denominator with the factor 600 in the numerator leaves

$$\frac{5 \cdot 7 \cdot 25}{7^2 \cdot 13} = \frac{5 \cdot 25}{7 \cdot 13} = \frac{125}{91}.$$

Thus

$$\frac{PQ}{PE} = \frac{125}{91}.$$

0.1.8 8. Answer

$a = 125$, $b = 91$ are coprime positive integers, so

$$a + b = 125 + 91 = 216.$$

$$\boxed{216}$$